MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION ANSWER KEY March 2026

Fourth Semester

DSE-MG4DSEECT203-SINGLE BOARD COMPUTERS FOR IOT APPLICATIONS

(2024 ADMISSION ONWARDS)

PART A

MCQ

Each question carries 1 marks

1. b) Connect Arduino to computer
2. b) Floating input
3. c) Transmitter and receiver
4. c) Embedded C
5. b) Mathematical calculation
6. a) Arm processor
7. b) Device properties and data model
8. b) Multiplying length and breadth
9. a) Sensors and actuators
10. b) Arduino IDE
11. b) Assignes a value
12. b) Output device
13. b) Linux
14. b) Instant data processing
15. a) Computer vision applications
16. c) Maxwell

PART B

Short Answer Questions

Answer any 6 questions

Each question carries 5 marks

17. (Definition-2.5 marks, Explanation-2.5 marks)

Arduino board is a small microcontroller board, which is a small circuit(the board) that contains a whole computer on a small chip(the microcontroller).Arduino Uno consists of 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header and a reset button.

It consists of 14 digital inputs/output pins, each of which provide or take up 40mA current. Some of them have special functions like pins 0 and 1, which act as Rx and Tx respectively, for serial communication, pins 2 and 3-which are external interrupts, pins 3,5,6,9,11 which provides pwm output and pin 13 where LED is connected.

18. (Definition-2.5 marks, Features-1.25 marks each)

➔Blynk Cloud serves as the backbone for the Blynk ecosystem, offering a cloud-based solution that facilitates the communication between devices and the Blynk mobile app.

➔ It enables users to build connected projects without the need for extensive backend infrastructure or server management.

➔ At its core, Blynk Cloud simplifies the process of connecting hardware to the internet, making IoT development accessible to a broad audience.

19. (Definition-2 marks, Characteristics-3 marks)

The Internet of Things (IoT) refers to a network of interconnected physical objects such as devices, machines, vehicles, or people embedded with sensors, software, and unique identifiers that enable them to collect, exchange, and process data over a network without requiring direct human-to-human or human-to-computer interaction.

Characteristics of IoT

INTERCONNECTIVITY- With regards to IoT, anything can be interconnected to the global information and communication infrastructure.

HETEROGENEITY- The devices in IoT are heterogeneous based on different hardware platforms, applications and networks. They can interact with other devices through different networks.

SAFETY- IoT must be designed with safety in mind. This includes safety of our personal data and the safety of our physical devices.

Naming & Addressing- The names and addresses of devices connected to Internet of Things should be unique.

20. (Definition-2 marks, Features-3 marks)

Raspbian, officially rebranded as Raspberry Pi OS, is the official, free operating system developed by Raspberry Pi Holdings specifically for their single-board computers. Built on top of Debian Linux, it is optimized to run smoothly on the low-power hardware of the Raspberry Pi.

Optimized for the Pi: It is specifically compiled to take full advantage of the Raspberry Pi's processor and hardware features, offering the best out-of-the-box performance.
Pre-packed Software: It comes with a massive suite of pre-installed educational and programming tools (like Python, Scratch, and Thonny) as well as everyday software like the Chromium/Firefox browsers and LibreOffice.
User-Friendly Desktop: It features a lightweight, user-friendly desktop environment (labwc / Wayland-based) that makes it feel familiar to Windows or macOS users.

21. (Definition-2.5 marks, Explanation-1.25 marks each)

RASPBERRY PI MODELS

There have been many generations of the Raspberry Pi line: from Pi 1 to 4, and even a Pi 400.

There has generally been a Model A and a Model B of most generations.

Model A has been a less expensive variant, and tends to have reduced RAM and fewer ports (such as USB and Ethernet). The Pi Zero is a spinoff of the original (Pi 1) generation, made even smaller and cheaper.

Raspberry pi 4 model b

- This model has just been released.
- This model has broken the mold of connector positioning on the Raspberry Pi that has remained the same since the release of the Raspberry Pi B+ in 2014.

The Pi Zero W is roughly half the size of the Pi 3 B or Pi 4 B, and it has a single micro-USB socket for communication, and a second one for power.

- The Pi Zero W also saves space by using a mini-HDMI socket and micro-USB on-the-go socket.
- If you want to connect a keyboard, monitor and mouse to a Pi Zero, you will need adapters for both the USB and the HDMI port before you can connect standard peripherals.

22. (Definition-2 marks, Explanation-3 marks)

SD Card Slot

What it is: Slot for the MicroSD card**.**

Function: Stores the operating system, programs, and data. The Pi boots from this card.

Location: On the underside or edge of the Raspberry Pi board.

Tip: Always use a Class 10 MicroSD card for faster performance**.**

23. (Definition-2 marks, Applications-1.5 marks each)

Camera interfacing with OpenCV enables real-time image processing for tasks like face recognition and object tracking, making it useful in security, automation, and computer vision applications.

- **Face Detection and Recognition**
  - OpenCV can capture live video from a camera and detect human faces in real time.
  - It is widely used in security systems, attendance monitoring, and smartphone authentication.
- **Object Detection and Tracking**
  - OpenCV processes camera images to identify and track objects such as vehicles, people, or products.
  - Applications include surveillance systems, autonomous robots, and traffic monitoring.

24. (Definition-2 marks, Applications-1.5 marks each)

Banana Pi is a versatile single-board computer used in applications such as home automation and multimedia systems due to its processing power, connectivity, and low cost.

- **Home Automation System**
  - Banana Pi can be used to control and monitor home appliances such as lights, fans, and security devices through the internet.
  - It acts as a central controller for smart home applications.
- **Media Center and Entertainment System**
  - Banana Pi can function as a multimedia server to play videos, music, and online content on TVs and monitors.
  - It supports high-definition video playback and streaming.

PART C

Essay Questions

Answer any 2 questions

Each question carries 12 marks

25. (Definition-3 marks, Features-3 marks each)

Feature	Arduino UNO	Arduino Mega
Memory	32 KB Flash Memory, 2 KB SRAM	256 KB Flash Memory, 8 KB SRAM
I/O Pins	14 Digital I/O Pins (6 PWM), 6 Analog Inputs	54 Digital I/O Pins (15 PWM), 16 Analog Inputs
Applications	Suitable for simple projects such as LED control, basic sensors, and small automation systems	Suitable for complex projects such as robotics, 3D printers, industrial automation, and projects requiring many sensors and actuators

26. **(Specifications-2 marks each)**

Different Raspberry Pi Models and Their Specifications

1. Raspberry Pi 1 Model B

- Processor: 700 MHz Single-Core ARM11
- RAM: 512 MB
- USB Ports: 2
- Ethernet: 10/100 Mbps
- Application: Basic programming and educational projects

2. Raspberry Pi 2 Model B

- Processor: 900 MHz Quad-Core ARM Cortex-A7
- RAM: 1 GB
- USB Ports: 4
- Ethernet: 10/100 Mbps
- Application: Home automation and embedded systems

3. Raspberry Pi 3 Model B

- Processor: 1.2 GHz Quad-Core ARM Cortex-A53
- RAM: 1 GB

- Connectivity: Wi-Fi and Bluetooth built-in
- USB Ports: 4
- Application: IoT projects, media centers, and robotics

4. Raspberry Pi 4 Model B

- Processor: 1.5 GHz Quad-Core ARM Cortex-A72
- RAM: 2 GB, 4 GB, or 8 GB
- Connectivity: Gigabit Ethernet, Wi-Fi, Bluetooth 5.0
- USB Ports: 2 USB 3.0 + 2 USB 2.0
- Application: Desktop computing, AI, and advanced IoT applications

5. Raspberry Pi Zero W

- Processor: 1 GHz Single-Core ARM11
- RAM: 512 MB
- Connectivity: Wi-Fi and Bluetooth
- Size: Very compact and low power
- Application: Wearable devices and portable embedded systems

6. Raspberry Pi 5

- Processor: 2.4 GHz Quad-Core ARM Cortex-A76
- RAM: 4 GB or 8 GB
- Connectivity: Dual-band Wi-Fi, Bluetooth 5.0, Gigabit Ethernet
- USB Ports: 2 USB 3.0 + 2 USB 2.0
- Application: Edge AI, machine learning, industrial automation, and high-performance computing

27. (Definition-2 marks, Importance-2 marks each)

Linux commands play a vital role in operating and managing a Raspberry Pi because the Raspberry Pi OS is based on Linux. They allow users to efficiently control the system, manage files, configure hardware and troubleshoot issues.

- **File and Directory Management**
    - Commands such as `ls`, `cd`, `mkdir`, and `rm` help users create, access, organize, and delete files and folders.
- **System Administration**
    - Commands like `sudo`, `passwd`, and `shutdown` are used for user management, system control, and security.
- **Software Installation and Updates**
    - Commands such as `apt update` and `apt install` enable users to install, update, and manage software packages.

- **Hardware and Network Configuration**
    - Linux commands help configure GPIO pins, network settings, Wi-Fi, Bluetooth, and connected devices.

- **Troubleshooting and Monitoring**
    - Commands like `top`, `df`, `free`, and `ping` are used to monitor system performance and diagnose problems.

28. (Definition-2 marks, Applications-2 marks each)

Tinker Board is a powerful single-board computer used for developing embedded and IoT applications due to its high processing capability, GPIO support, and connectivity features.

Applications:

1. Home Automation

- Used to control smart devices such as lights, fans, and security systems.
- Supports remote monitoring and control through the Internet.

2. IoT Gateway

- Collects data from sensors and transfers it to cloud platforms for analysis and monitoring.
- Acts as a bridge between IoT devices and servers.

3. Robotics and Automation

- Controls motors, sensors, and robotic components.
- Used in autonomous robots and industrial automation projects.

4. Smart Surveillance Systems

- Interfaces with cameras for video monitoring, motion detection, and security applications.

- Supports real-time image processing.

5. Industrial Monitoring

- Monitors temperature, pressure, humidity, and other industrial parameters.
- Enables predictive maintenance and remote diagnostics.

6. Multimedia and Edge Computing

- Used for media streaming, digital signage, and edge computing applications.
- Processes data locally before sending it to the cloud.